Colostrum contains an interferon like substance which has strong antiviral activity. It contains a B12 binding protein making it unavailable for the growth of ***E-coli*** and other bacteria. It also contains antibodies against viral infection.

Advantages of breast feeding

Breast feeding is the simple and best method of feeding and has the following advantages

1. Nutritional factor

The composition of human milk is best suited for infants. In human milk the protein content is lower but the content of carbohydrate, namely lactose is higher. The fat content is comparatively less. The protein is present as lactalbumin which is better digested than the protein in cow's milk. Lactose provides natural sweetness and also helps in absorption of calcium and iron. Fat though less is highly emulsified and therefore better digested. When compared to animal milk, breast milk provides higher amount of vitamin C. Similarly calcium in breast milk though less when compared to cow's milk is better absorbed by the infant. The composition of human milk is best suited for infants. The table 8 below shows the comparison of Human milk and cow's milk.

2. Hormones and growth factors:

Breast milk is a rich source of hormones like Thyroid Stimulating Hormone (TSH), thyroxin, insulin and prolactin. It also contains growth regulating factors, growth promoters and growth modulators.

Table 8 Comparison Of Human Milk and Cow's Milk

Nutrient per 100 ml	Human milk	Cow's Milk
Water (g)	88	87.5
Energy (k cal)	65	67
Protein (g)	1.1	3.2
Carbohydrate (g)	7.4	4.4
Fat (g)	3.4	4.1
Calcium (mg)	28	120
Phosphorus (mg)	11	90
Iron (mg)	– –	0.2
Carotene (µg)	137	174
Thiamine (mg)	0.02	0.05
Riboflavin (mg)	0.02	0.19
Vitamin C (mg)	3	2
Caseinogen lactalbumin ratio	1.2	3.1

3. Immunological factors:

The following factors in breast milk provide passive immunity.

a. **Macrophages:** They can digest bacteria and also develop immunity against infectious diseases.

b. **Lymphocytes:** Lymphocytes produce antiviral substances like interferon.

c. **Lactoferrin:** It is an iron binding protein that inhibits the growth of *E.coli* and other bacteria.

d. **Enzymes:** Breast milk also supplies enzymes like lipase, amylase and lactoperoxidase which increase digestibility and also destroy the harmful microorganisms.

e. **Immunoglobulin:** They are defensive proteins which include all types of antibodies.

DO YOU KNOW?

August 1-7 is the World Breast Feeding week

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